

Unequal Job Opportunities and Well-Being Inequality

A Latent-Jobs Structural Decomposition

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DSFEM PhD Workshop

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Introduction: three pillars

1. Normative welfare

What should individuals
be compensated for? $\implies$

Why does the *source*
of inequality matter?

2. Latent job opportunities

Observed choice
 $\neq$

observed choice set

Preferences +
opportunity environment

$\implies$

3. Structural decomposition

How much inequality
is associated with
each modelled channel?

Shapley attribution

Normative welfare $\longrightarrow$ RURO $\longrightarrow$ Shapley decomposition

The data show the realised choice. They do not show the menu behind it.

Inequality of opportunity: an established motivation

How much inequality is associated with circumstances outside individual responsibility?

$\approx 20\%$

Italy

Checchi & Peragine (2010)

10–37%

Brazil

Bourguignon et al. (2007)

2–34%

Cross-country evidence

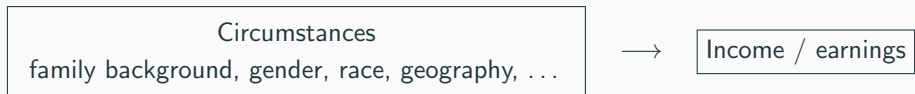
Brunori et al. (2013)

Illustrative magnitudes, not a harmonised comparison: outcomes, circumstances, samples and inequality indices differ.

Circumstances can account for an economically important share of inequality.

From circumstances to job opportunities

Reduced-form inequality-of-opportunity perspective



But through which labour-market mechanism?



Main question

How much inequality in money-metric well-being is associated with unequal job opportunities rather than heterogeneous preferences, once labour supply is modelled as choice among latent jobs?

1. Behaviour

Do observed labour-supply choices reflect preferences alone, or also heterogeneous latent job opportunities?

2. Welfare

How do conclusions differ between the ex-post attained bundle and the ex-ante opportunity prospect?

3. Decomposition

How much welfare inequality is associated with preferences, local access, and earning opportunities?

Next extension: household resources, needs and composition

Preview of preliminary results

Current results are preliminary and remain subject to ongoing revision.

1. Welfare perspective changes the magnitude

	Single adults	Couples
Ex-ante prospect	20.3%	7.9%
Attained bundle	1.9%	3.5%

Local access + earning opportunities as a share of each perspective's baseline Gini, equivalised.

2. For singles, the dominant channel changes

Ex ante: $\underbrace{15.5\%}_{\text{local access}} > \underbrace{4.8\%}_{\text{earnings}}$

Attained bundle: earnings > local access

Well-being: compensation versus responsibility

Compensation

Unequal productive circumstances
should not automatically become
unequal well-being

Responsibility

Preference-based choices
should not automatically
be compensated

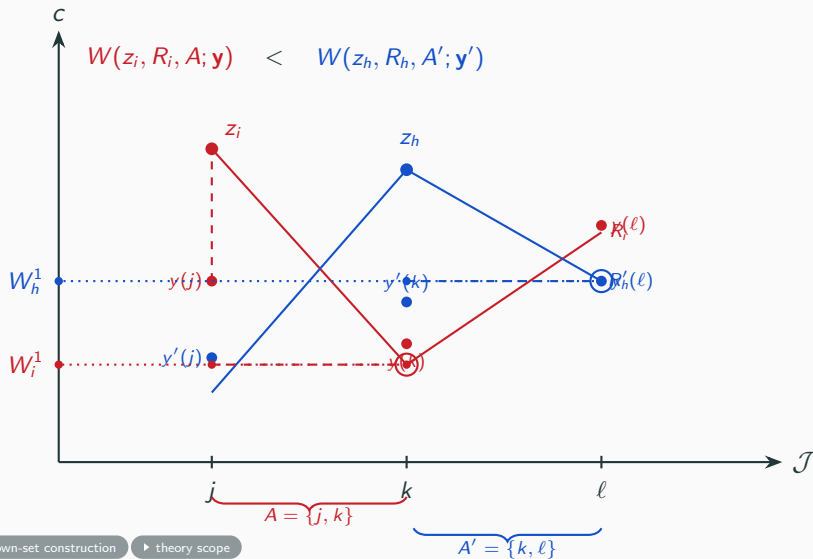
Compensation vs. Responsibility

$\nexists W : \text{Full Compensation} \wedge \text{Full Responsibility}$

[Fleurbaey & Maniquet, *SCW* 2005]

Full Comp. + weak Resp. **or** Full Resp. + weak Comp.

The measures



From the theoretical ability set to latent opportunities

$$W(z, R, A; \mathbf{y}) = w \Leftrightarrow z \not\prec_R \max\{B\}$$

where

$$z' = (c', j') \in B \Leftrightarrow j' \in A \text{ and } c' = w.$$

Common consumption $\times$ own preferences $\times$ own feasible jobs

From the theoretical ability set to latent opportunities

$$W(z, R, A; \mathbf{y}) = w \Leftrightarrow z \in \max_R \{B\}$$

where

$$z' = (c', j') \in B \Leftrightarrow j' \in A \text{ and } c' = w.$$

Common consumption $\times$ **own preferences** $\times$ **own feasible jobs**

$$A_i \longrightarrow \hat{g}_i(j)$$

Theory

feasible ability set

Empirics

relative opportunity density

The exact historical menu is not observed.

What consumption while staying home is equivalent to the attained bundle?

$$B(x; \theta) = \begin{cases} \frac{x^\theta - 1}{\theta}, & \theta \neq 0, \\ \log x, & \theta = 0. \end{cases}$$

$$v_i(j) = \beta_\ell B\left(\frac{\ell_i(j)}{\lambda_\ell}; \theta_\ell\right) + \beta_c B\left(\frac{C_i(j)}{\lambda_c}; \theta_c\right), \quad \text{with } \beta_\ell B\left(\frac{\ell_i(j)}{\lambda_\ell}; \theta_\ell\right) = L_i(j).$$

$$W_i^{\text{ATT}} = \lambda_c B^{-1} \left[B\left(\frac{C_i^{\text{obs}}}{\lambda_c}; \theta_c\right) + \frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c}; \theta_c \right].$$

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$$W_i^{\text{ATT}} = \lambda_c B^{-1} \left[B\left(\frac{C_i^{\text{obs}}}{\lambda_c}; \theta_c\right) + \frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c}; \theta_c \right].$$

$$\theta_c = 0 \quad \Rightarrow \quad \boxed{W_i^{\text{ATT}} = C_i^{\text{obs}} \exp \left\{ \frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c} \right\}}$$

Conditional on the attained bundle, $\hat{g}_i$ does not enter welfare directly.

Ex-ante prospect welfare

What constant consumption over the household's own opportunity environment is equivalent to its actual prospect?

$$J_i = \int e^{L_i(j)} \left(\frac{C_i(j)}{\lambda_c} \right)^{\beta_c} \widehat{g}_i(j) d\nu(j),$$

$$H_i = \int e^{L_i(j)} \widehat{g}_i(j) d\nu(j),$$

$$W_i^{\text{EA}} = \lambda_c \exp \left\{ \frac{\log J_i - \log H_i}{\beta_c} \right\}.$$

J_i
actual prospect

H_i
same opportunity env.,
equal cons.

W_i^{EA}
flat-cons. equiv.

actual prospect $\iff$ same opportunities + flat consumption
 J_i H_i, W_i^{EA}

RURO estimation: preferences versus opportunities

Observed choices reflect both how much a job is valued and how available that type of job is.

$$P_i(j) = \frac{\exp\{v_i(j)\} g_i(j)}{\int \exp\{v_i(k)\} g_i(k) d\nu(k)}$$

$$V_{ij} = v_i(j) + \log g_i(j) - \log q_{ij}, \quad \Pr(y_i | \mathcal{C}_i) = \frac{n_{y_i} \exp V_{iy_i}}{\sum_{s \in \mathcal{C}_i} \exp V_{is}}.$$

$|\mathcal{C}_i| = 101$: 100 proposal draws + the observed package.

$-\log q_{ij}$ corrects for the proposal sampling.

RURO estimation: preferences versus opportunities

Opportunity specification

$$g_i(o) = 1$$

$$g_i(j \mid E = 1) = g_i^E g_i^H(h) g_i^{Occ}(k) g_i^W(w \mid k).$$

- **Access:** local unemployment, region, urbanisation
- **Hours:** five hours bands
- **Occupation:** occupation-group mass
- **Wage offers:** log-normal; location shifts with education, experience and occupation

What separates preferences from opportunities?

1. **Functional-form separation**
preferences smooth in hours; opportunity density is banded
2. **Excluded access shifters**
local unemployment, region and urbanisation shift access but not utility
3. **Offer structure**
wage offers independent of offered hours conditional on occupation

Identification is structural and parametric.

These are maintained restrictions, not a causal design.

Capéau, Decoster & Dekkers (2016); Dagsvik & Jia (2016)

Observed households

EU-SILC France

collection 2016, incomes 2015

- demographics and household composition
- employment status, hours and earnings
- education and occupation
- geography and survey weights

1,540 single-adult households

2,223 couple households

Latent job packages

$$j = (e, k, h, w)$$

- employment / non-employment
- occupation
- weekly hours
- hourly wage

$$h \in [5, 70], \quad w \in [2, 590]$$

The observed job is retained; alternative jobs are sampled from the proposal distribution.

Data, simulated jobs and EUROMOD (cont.)



French 2015 policy system

$C_i(j)$ = disposable income generated by job j for household i .

Income tax + social contributions + transfers $\Rightarrow$ exact nonlinear budget mapping

EUROMOD prices the alternatives; RURO models the choice.

EUROMOD is run before estimation, and the resulting disposable incomes are then used in the likelihood.

Sutherland & Figari (2013)

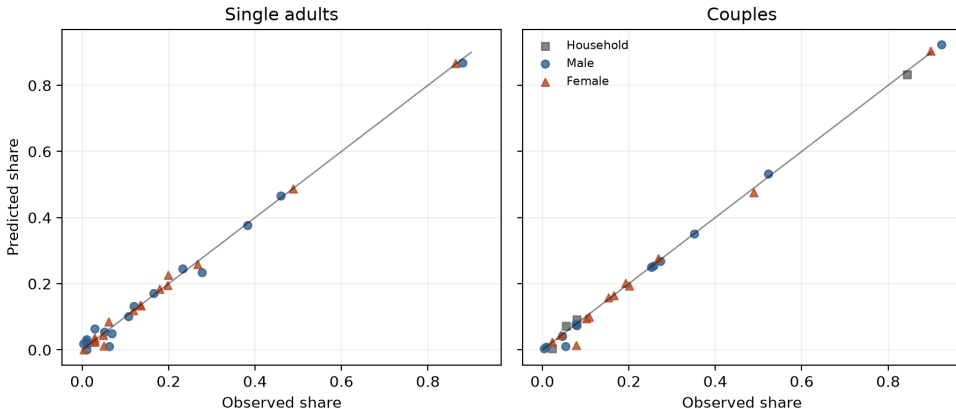
Does the model reproduce observed labour-market choices?

► fit by margin

► extensive margin

Population predictions: integrating over latent opportunities and taste shocks, not in-sample fitted choice probabilities.

Corrected population moments: observed versus predicted



Does the model reproduce observed labour-market choices? (cont.)

Single adults

Mean absolute deviation
across reported moments: **0.0140**

Couples

Mean absolute deviation
across reported moments: **0.0119**

Participation and occupation fit closely in aggregate; hours fit is more uneven.

Does heterogeneous opportunity structure matter?

Single-adult population fit:

0.0140
household-specific
opportunities

vs.

0.0273
common opportunity A

0.0264
common opportunity B

Main remaining fit issue: underprediction of the exact 37-hour mass point.

From individual well-being to welfare inequality

For each welfare perspective, I obtain one money-metric well-being level for every household.

Primary reporting scale

$$\tilde{M}_i^p = \frac{M_i^p}{e_i}, \quad p \in \{\text{ATT}, \text{EA}\}$$

Modified OECD equivalence scale
household survey weights

Singles and couples reported separately

Raw household welfare is retained as a secondary scale.

Inequality statistic: weighted Gini

Let normalized household weights satisfy

$$\sum_i \omega_i = 1, \quad \mu_M = \sum_i \omega_i M_i.$$

$$G(M) = \frac{\sum_i \sum_j \omega_i \omega_j |M_i - M_j|}{2\mu_M}$$

$G = 0$: equal well-being

larger G : more dispersed well-being

Structural decomposition of welfare inequality

To attribute inequality to a channel, I equalise that channel inside the model and recompute welfare.

P: Preferences

preference shifters

→ common reference

A: Local access

local unemployment,
region, urbanisation

→ common reference

B: Earnings

wage-offer location

→ common reference

D: Resources / needs

household composition
and resources

held fixed

$$x_i = (x_i^P, x_i^A, x_i^B, d_i)$$

$$T_P x_i = (\bar{x}^P, x_i^A, x_i^B, d_i),$$

$$T_A x_i = (x_i^P, \bar{x}^A, x_i^B, d_i),$$

$$T_B x_i = (x_i^P, x_i^A, \bar{x}^B, d_i).$$

Structural decomposition of welfare inequality (cont.)

$$I^P(S) = \mathcal{G} \left(\{M_i^P(T_S x_i)\}_{i=1}^N \right)$$

$$v^P(S) = I^P(\emptyset) - I^P(S)$$

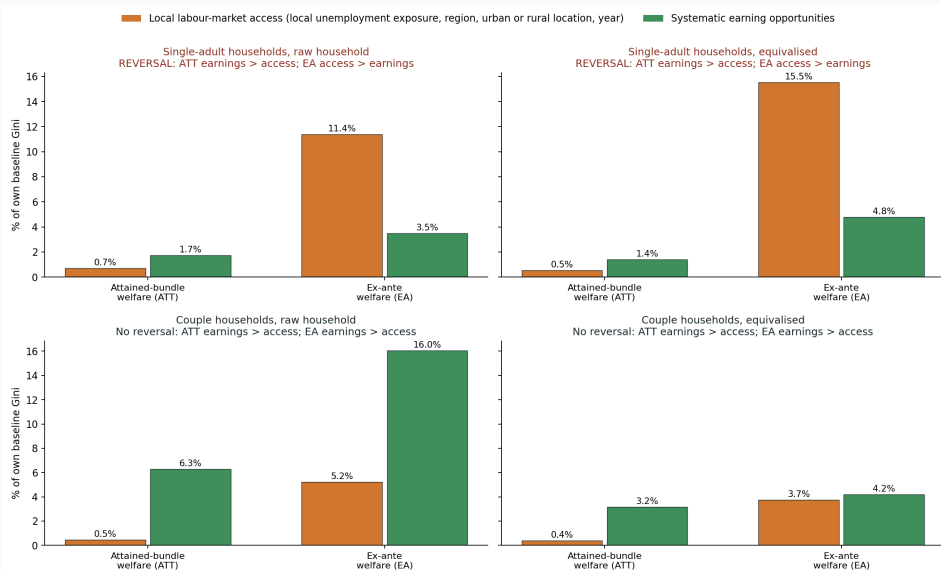
Baseline Gini – counterfactual Gini = inequality removed by equalising coalition S

$$\phi_k^P = \sum_{S \subseteq \{P, A, B\} \setminus \{k\}} \frac{|S|!(3 - |S| - 1)!}{3!} [v^P(S \cup \{k\}) - v^P(S)].$$

$$\phi_P^P + \phi_A^P + \phi_B^P = I^P(\emptyset) - I^P(\{P, A, B\}) \equiv \Delta I^P.$$

Average each channel's marginal contribution over all $3! = 6$ orders.

How much is associated with job opportunities?

[▶ full-size figure](#)[▶ EA channels](#)[▶ ATT channels](#)

How much is associated with job opportunities? (cont.)

Access + earning opportunities, equivalised:

Single adults

Ex-ante prospect: **20.3%**

Attained bundle: **1.9%**

Couples

Ex-ante prospect: **7.9%**

Attained bundle: **3.5%**

Singles: the ranking of the opportunity channels reverses

EA: access > earnings

ATT: earnings > access

Shares are relative to each perspective's own baseline Gini. The ATT–EA comparison is not a pure welfare-definition experiment: aggregation and numerical support also differ.

How should I interpret these numbers?

▶ ATT channels

▶ EA channels

▶ composition within P-A-B

▶ aggregation orders

Attained-bundle welfare

Opportunity environment



attained job



consumption + leisure



well-being

Once the attained bundle is fixed,
unchosen opportunities disappear
from the welfare evaluation.

Ex-ante prospect welfare

Whole job environment



availability $\times$ value of every job



value of the prospect



well-being

Jobs that are not ultimately chosen
still enter the welfare evaluation
through their opportunity weight.

Earning opportunities $>$ access

For singles: access $>$ earnings

How should I interpret these numbers? (cont.)

Three qualifications

1. **Partial opportunity accounting:** A and B are selected opportunity channels, not the complete opportunity environment.
2. **Scale matters:** equivalisation changes the quantitative attribution materially, especially for couples.
3. **Structural accounting, not causal shares:** parameter uncertainty has not yet been propagated through the welfare and Shapley calculations.

ATT–EA differences are consistent with the two welfare perspectives valuing different objects, but aggregation and numerical support also differ.

1. Latent job opportunities matter empirically

Allowing household-specific opportunity heterogeneity improves the representation of observed labour-supply choices relative to common-opportunity benchmarks.

2. The welfare perspective matters for the decomposition

On the equivalised scale, access + earning opportunities account for about **20.3%** of ex-ante welfare inequality for singles and **7.9%** for couples, compared with only **1.9%** and **3.5%** under attained-bundle welfare.

3. The opportunity mechanism matters

For singles, local access dominates earning opportunities under ex-ante welfare, while the ordering reverses under attained-bundle welfare.

For couples, the ex-ante contributions of access and earnings are close.

Conclusion (cont.)

Looking only at the job that was attained can miss inequality carried by the opportunity environment behind that choice.

Next: extend the decomposition to household resources, needs and composition; propagate parameter uncertainty; strengthen the attained-bundle numerical implementation.

Preliminary structural accounting results; not causal attribution.

Thank you

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Appendix map

Literature	EOp benchmarks	Literature	
Theory	The measures	Own-set construction	Companion theory scope
Model	Utility specification	Opportunity specification	Likelihood
	Identification	Observational equivalence	
Estimation	Preference parameters	Access parameters	Hours and occupation
	Wage offers	Common opportunities	Curvature and inference
	Parameter uncertainty		
Fit	Extensive margin	Fit by margin	37-hour point
	Matched households	Marginal rate of substitution	
Welfare	ATT derivation	ATT numerical design	EA derivation
	EA validation	Prospect reference	Two aggregation orders
	Equivalisation		
Decomposition	Central figure	EA, Gini points	ATT, Gini points
	EA, % of baseline	ATT, % of baseline	Composition within P-A-B
	Shapley vs one-factor	Outside P-A-B	Robustness
Data	Sample selection	EUROMOD pricing	Numerical design

Backup: inequality-of-opportunity benchmarks

	share	outcome	data	measure	source
Italy	19.5%	gross earnings	SHIW 1993–2000	MLD; parental education	Checchi & Peragine (2010), p. 445
Brazil, urban	10–37%	male hourly earnings	PNAD 1996	Theil; five circumstances	Bourguignon et al. (2007), abstract
France	12.9%	post-tax earnings	EU-SILC 2005	MLD; five circumstances	Brunori et al. (2013), Table 1

These benchmarks

shares of earnings inequality attributed to observed circumstances

This paper

a share of welfare inequality from estimated latent opportunity components, resources and needs held fixed

Checchi & Peragine (2010), Inequality of opportunity in Italy, *Journal of Economic Inequality*. Bourguignon, Ferreira & Menéndez (2007), Inequality of opportunity in Brazil, *Review of Income and Wealth*. Brunori, Ferreira & Peragine (2013), Inequality of opportunity, income inequality and economic mobility, IZA DP; its France row reports Checchi, Peragine & Serlenga.

Choice sets and restrictions

Aaberge, Colombino & Wennemo (2009): alternative representations of the choice set

Beffy, Blundell, Bozio, Laroque & Tô (2019): labour supply with restricted choices

Latent jobs and welfare

Aaberge, Dagsvik & Strøm;
Dagsvik & Strøm; Dagsvik & Jia;
Capéau, Decoster & Dekkers
Fleurbaey & Maniquet; Decoster
& Haan; Bargain et al.; Jacquet,
Jia & Thoresen

Inequality of opportunity and decomposition

Roemer; Checchi & Peragine;
Bourguignon, Ferreira &
Menéndez; Brunori, Ferreira &
Peragine

Shorrocks; Sastre & Trannoy;
Creedy & Hérault; Mühlhan

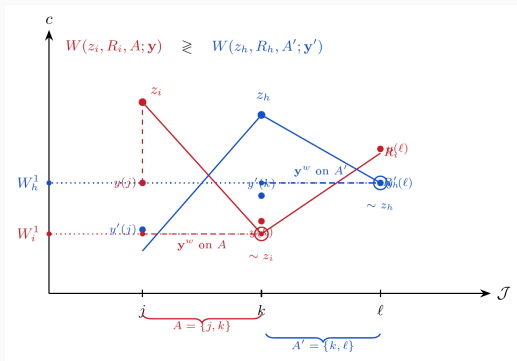
Backup: the measures

measure	definition	characterised by
W^1 Equal Pay	$W = w \Leftrightarrow z \text{ I } \max_R \{B\}, \quad B = \{(w, j') : j' \in A\}$	Resp. for Equal Pay + Ind. of y
W^2	$W = w \Leftrightarrow z \text{ I } \max_R \{B\},$ $B = \{(y(j') - t, j') : j' \in A\}, \quad w = \max_A y - t$	Full Resp. + Comp. for R^h
W^3 Laissez-Faire	$W = w \Leftrightarrow z \text{ I } \max_R \{B\},$ $B = \{(y(j') + w, j') : j' \in A\}$	Full Resp. + Equal Well-Being at Laissez-Faire
W^4 Staying-Home Equivalent	$W = w \Leftrightarrow z \text{ I } (w, o)$	Full Comp. + Ind. of Prefs. over Infeasible Jobs
W^5 Reference Ability	$W = w \Leftrightarrow z \text{ I } \max_R \{B\},$ $B = \{(y(j') + w, j') : j' \in \bar{A}\}$	Ind. of A + Resp. for Reference Abilities; $\bar{A} = \mathcal{J}$: Resp. when Preferred Job Possible
W^6 Min of Equal Pay	W equal whenever $z \text{ I } \arg \max_{(\mathcal{J}, y^w)} R$ and $z' \text{ I}' \arg \max_{(\mathcal{J}, y^w)} R'$	Full Comp. + Weak Resp.

maintained throughout: Representation, Job Duplication Invariance, Job Neutrality, continuity and monotonicity in consumption

· Haydar & Maniquet, work in progress; proofs in the companion theory talk

Backup: own-set equal-consumption construction



Own-set equal-consumption equivalents: each individual gets a common consumption on every job in their own ability set; the level at which the preferred job is indifferent to the attained bundle is their money metric, comparable across individuals. Adapted from Haydar and Maniquet (2026), work in progress.

What the companion theory does and does not cover

Covered

The deterministic own-set equal-consumption construction is characterised axiomatically in the companion paper.

Not covered

The stochastic extension to a prospect is this paper's own step and is not characterised there.

The asymmetry

The attained-bundle measure uses systematic utility only, while the prospect measure's expected-maximum construction makes the taste component relevant to the prospect's value.

Whether that asymmetry is the right normative choice is open.

Backup: utility specification

$$v_i(j) = L_i(j) + \beta_c \log\left(\frac{C_i(j)}{\lambda_c}\right), \quad L_i(j) = \beta_\ell^g(\mathbf{x}_i) \mathcal{B}(\tilde{\ell}_i(j); \theta_\ell^g)$$

$$\tilde{\ell}_i(j) = \frac{80 - h}{10}, \quad \mathcal{B}(z; \theta) = \frac{z^\theta - 1}{\theta}$$

$\beta_\ell^g(\mathbf{x}_i)$

intercept, centred age and its
square (units of 10 years)

k_i

children under 20, women only

couples

sum of the two spouses' leisure
terms

current specification: $\theta_c = 0$, so $\beta_c B(C/\lambda_c; \theta_c) = \beta_c \log(C/\lambda_c)$

Backup: opportunity specification

$$g_i(o) = 1, \quad g_i(j) = g_i^E g_i^H(h) g_i^{\text{Occ}}(k) g_i^W(w | k) \quad \text{for market jobs } j \neq o$$

access

g_i^E : local unemployment exposure, region, urban or rural location, year

hours

g_i^H : elevated bands over a residual set

occupation

g_i^{Occ} : sex-specific group masses

wage offers

g_i^W : log-normal, truncated to $[2, 590]$

wage-offer location $\mu_i(k)$: education, potential experience and its square, occupation

Backup: sampled-alternative likelihood

$$V_{ij} = v_i(j) + \log g_i(j) - \log q_{ij}$$

$$\Pr(y \mid \mathcal{C}_i) = \frac{n_y \exp V_{iy}}{\sum_{s \in \mathcal{C}_i} \exp V_{is}}$$

q_{ij}

proposal density, fitted out of fold

$\mathcal{C}_i$

observed job and 100 draws: 101
rows

n_y

multiplicity of a repeated draw

Backup: what separates preferences from opportunities?

Functional form

smooth preferences in hours;
banded hours opportunities

Exclusion

local unemployment exposure and
geography shift access, not utility

Independence

wage offers independent of hours,
given occupation

maintained, not tested · not a causal design

Backup: observational equivalence

Without restrictions

a taste against certain hours and a scarcity of jobs at those hours predict the same choices

With restrictions

smooth tastes, banded availability and excluded access shifters move different parts of the choice distribution

If availability heterogeneity is omitted, some of its effects can be absorbed by the estimated utility component.

maintained restrictions, not tested · regional variation is not a causal design

Backup: preference parameters

	Single men	Single women	Couple men	Couple women
leisure weight, intercept	8.5199 (3.5480) <code>beta_l0_sm</code>	5.8683 (2.1346) <code>beta_l0_sf</code>	3.9814 (0.7466) <code>beta_l0_m</code>	13.1029 (4.7760) <code>beta_l0_f</code>
leisure weight, age	1.4820 (1.0236) <code>beta_l_age_sm</code>	0.0678 (0.4792) <code>beta_l_age_sf</code>	−0.0080 (0.0287) <code>beta_l_age_m</code>	−0.1228 (0.1260) <code>beta_l_age_f</code>
leisure weight, age squared	0.7783 (0.7732) <code>beta_l_age2_sm</code>	1.0000 (at bound) <code>beta_l_age2_sf</code>	0.0065 (0.0030) <code>beta_l_age2_m</code>	0.0073 (0.0103) <code>beta_l_age2_f</code>
leisure weight, children	—	0.1666 (0.4422) <code>beta_lnkids_sf</code>	—	−0.3055 (1.1472) <code>beta_lnkids_f</code>
leisure curvature θ_ℓ	−1.6263 (0.3301) <code>theta_l_sm</code>	−0.9274 (0.2161) <code>theta_l_sf</code>	−0.9761 (0.1357) <code>theta_l_m</code>	−1.6863 (0.2458) <code>theta_l_f</code>
consumption weight β_c	2.0387 (0.2917) <code>beta_c</code>		2.1017 (0.2939) <code>beta_c</code>	

cluster-robust standard errors; parameter identifiers in grey; the male children term in couples is structurally zero

Backup: access parameters

	identifier	Singles	Couples
employment constant	beta_E; beta_E_m, beta_E_f	−3.1735 (0.4053)	men −2.1233 (0.3195) women −2.8148 (0.3021)
local unemployment exposure	beta_E_gsur	−1.4422 (0.2356)	−1.1924 (0.1557)
densely populated	beta_E_drgur	−0.0288 (0.2209)	−0.1823 (0.1657)
intermediate density	beta_E_drgmd	0.0641 (0.2614)	−0.4174 (0.1861)
Region 2	beta_E_drgn2	−0.3780 (0.3292)	−0.1546 (0.2442)
Region 3	beta_E_drgn3	−0.1117 (0.3891)	0.0873 (0.2808)
Region 4	beta_E_drgn4	−0.8218 (0.3810)	0.0150 (0.3068)
Region 5	beta_E_drgn5	−0.5162 (0.3274)	−0.1455 (0.2555)
Region 6	beta_E_drgn6	−0.7222 (0.3528)	−0.2844 (0.2748)
Region 7	beta_E_drgn7	−0.5374 (0.3503)	−0.1396 (0.2651)
Region 8	beta_E_drgn8	−0.4512 (0.3401)	−0.1659 (0.2660)

references: the first region and thinly populated areas; couples share the shifters across spouses

Backup: hours and occupation opportunities

Hours bands					
	Singles		Couple men		Couple women
part-time lower	0.0298		−1.1094		−0.5173
	beta_h_pt1		beta_h_pt1_m		beta_h_pt1_f
part-time upper	0.7080		0.0835		1.1586
	beta_h_pt2		beta_h_pt2_m		beta_h_pt2_f
narrow full-time	2.0656		2.2888		2.0284
	beta_h_f35		beta_h_f35_m		beta_h_f35_f
full-time upper	1.9387		2.3801		1.6825
	beta_h_ft		beta_h_ft_m		beta_h_ft_f
long hours	−0.0834		0.6977		−0.3109
	beta_h_lh		beta_h_lh_m		beta_h_lh_f

Occupation masses				
	Singles		Couples	
	men	women	men	women
Occupation 2	−1.3737	0.1135	−1.5097	0.2074
	beta_occ_2_m	beta_occ_2_f	beta_occ_2_m	beta_occ_2_f
Occupation 3	−2.1096	−0.4060	−2.2532	−0.1804
	beta_occ_3_m	beta_occ_3_f	beta_occ_3_m	beta_occ_3_f
Occupation 4	−0.3259	0.5632	0.1594	0.8159
	beta_occ_4_m	beta_occ_4_f	beta_occ_4_m	beta_occ_4_f

estimates with identifiers; standard errors in the paper

Backup: wage-offer equation

	identifier	Singles	Couples
intercept	beta_w0	2.0138 (0.0561)	2.0480 (0.0335)
low education	beta_w_educL	0.0566 (0.0372)	-0.0433 (0.0217)
high education	beta_w_educH	0.1491 (0.0308)	0.1817 (0.0197)
potential experience	beta_w_pexp	0.2408 (0.0862)	0.5643 (0.0546)
experience squared	beta_w_pexp2	-0.0282 (0.0390)	-0.1598 (0.0243)
Occupation 2	delta_occ_2	-0.0347 (0.0399)	-0.0740 (0.0231)
Occupation 3	delta_occ_3	0.0622 (0.0383)	0.0394 (0.0228)
Occupation 4	delta_occ_4	0.2785 (0.0368)	0.2146 (0.0223)
log-wage dispersion σ	sigma	0.3815 (0.0133)	0.3631 (0.0072)

medium education is the reference; experience in units of 20 years

Backup: household-specific versus common opportunities

Single adults	free coordinates	criterion	population fit
household-specific opportunities	41	6253.463	0.0140
common opportunity A: common distribution	10	6403.974	0.0273
common opportunity B: common shape; employment and hours in utility	16	6395.108	0.0264

same households, sampled alternatives and criterion; not converted into a formal test

Single adults

free coordinates 41, interior 40

criterion 6253.463

smallest Hessian eigenvalue 0.1669

Couples

free coordinates 47, interior 47

criterion 10283.034

smallest Hessian eigenvalue 0.0771

cluster-robust standard errors on the household · eigenvalues on the interior block

Backup: parameter uncertainty

Estimated

cluster-robust standard errors for
every estimated coordinate,
clustered on the household

Not yet propagated

estimation uncertainty has not
been carried through either welfare
decomposition

Simulation checks

second seeds and repeat runs
measure numerical noise, not
sampling uncertainty

the decomposition shares are point estimates without confidence intervals

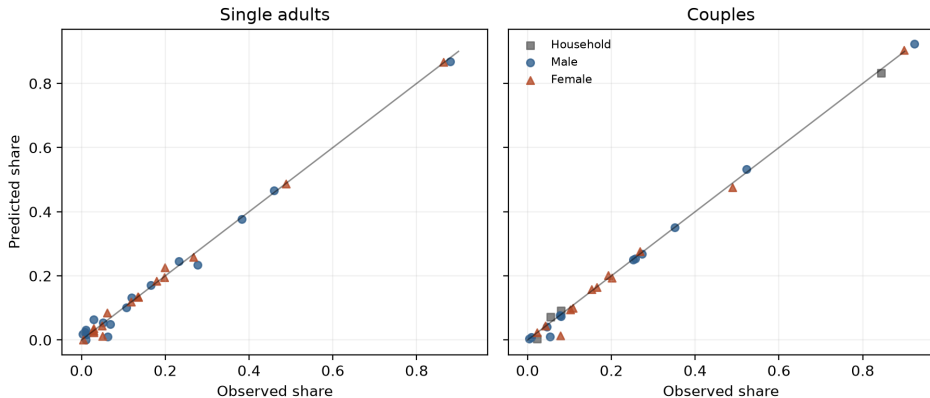
Backup: extensive-margin fit

	weighted accuracy	model-simulated range	integration ratio
single women	85.7%	80.0–85.9%	0.096
coupled women	89.8%	88.1–91.2%	0.049
single men	withheld	—	0.361
coupled men	withheld	—	0.353

withheld where numerical integration error is too large relative to sampling variation

Backup: fit by margin

Corrected population moments: observed versus predicted



Backup: the 37-hour mass point

single men

5.2

pp

single women

3.8

pp

coupled men

4.4

pp

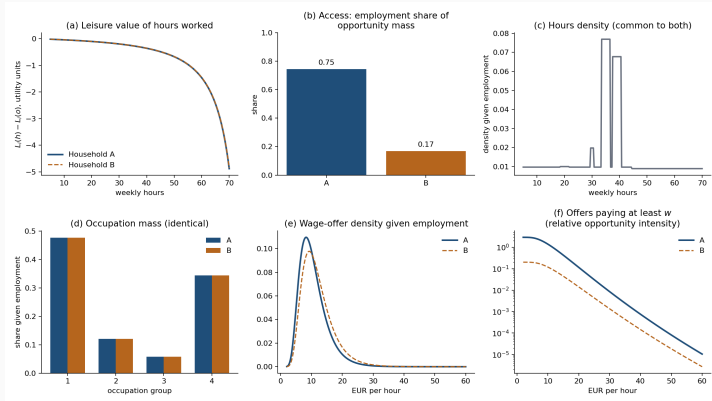
coupled women

6.5

pp

observed minus predicted share at the observed 37-hour mass point

Backup: matched households

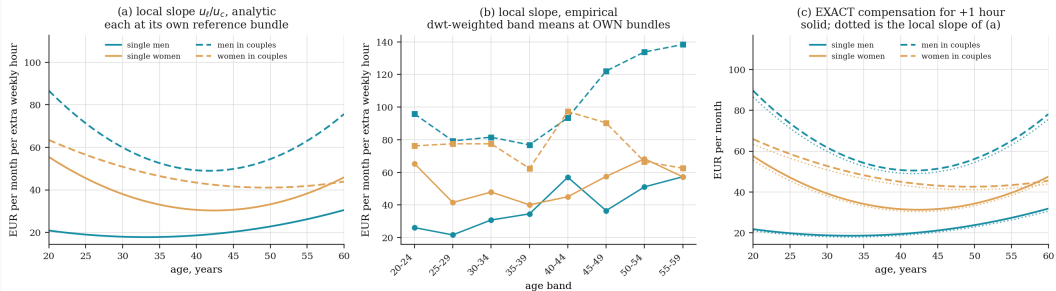


access: $A = 14.5 \times B$

wage-offer location: $B + 11.5$ log points

Backup: marginal rate of substitution

The local compensation u_l/u_c by age and sex, and the exact compensation for one more weekly hour



Backup: attained-bundle derivation and the staying-home equivalent

$$L_i(o) + \beta_c \log\left(\frac{W_i^{\text{ATT}}}{\lambda_c}\right) = L_i(j_i^{\text{obs}}) + \beta_c \log\left(\frac{C_i^{\text{obs}}}{\lambda_c}\right)$$

$$\implies W_i^{\text{ATT}} = C_i^{\text{obs}} \exp\left\{\frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c}\right\}$$

Reference job

the preferred job when every job pays the same:
here non-employment for every household

Staying-home equivalent

a verified property of this specification, not a
general theorem

on the estimated domain non-employment maximises L_i for every household, so the equal-pay preferred job is o
and the measure is the staying-home equivalent (W^4)

Backup: attained-bundle numerical design

Sample

the already-priced estimation
panel; no re-estimation and no
new pricing

Counterfactuals

equalised covariates set to
household-weighted means;
attained bundle simulated by
Gumbel-max with common
random numbers

Checks

replications at fixed estimates; an
independent second seed; exact
Shapley closure

short weekly hours are sparsely represented in the integration panel

Monte Carlo ranges are simulation spreads, not confidence intervals

Backup: ex-ante derivation

$$J_i = \int_{\Omega_i} e^{L_i(j)} \left(\frac{C_i(j)}{\lambda_c} \right)^{\beta_c} \widehat{g}_i(j) d\nu(j)$$

$$J_i^{\text{ref}}(m) = \left(\frac{m}{\lambda_c} \right)^{\beta_c} H_i, \quad H_i = \int_{\Omega_i} e^{L_i(j)} \widehat{g}_i(j) d\nu(j)$$

$$J_i^{\text{ref}}(W_i^{\text{EA}}) = J_i \quad \implies \quad W_i^{\text{EA}} = \lambda_c \exp \left\{ \frac{\log J_i - \log H_i}{\beta_c} \right\}$$

a common rescaling of $\widehat{g}_i$ cancels $\cdot \log J_i$: expected utility of the best reachable job, up to a constant

Backup: ex-ante numerical validation

Identities

constant-consumption identity

one-state identity

common intensity-rescaling invariance

flat-reference inversion

exact Shapley adding-up

Numerics

exact reference integral on the priced set

effective sample size and weight concentration

two-seed stability of welfare, Gini and Shapley

finite positive welfare for every household

independent reimplementation

all passed for both populations · a correct computation, not causal identification or parameter uncertainty

Backup: the prospect reference

$$W_i^{\text{EA}} = \left(\int C_i(j)^{\beta_c} \omega_i(j) d\nu(j) \right)^{1/\beta_c}, \quad \omega_i(j) = \frac{e^{L_i(j)} \hat{g}_i(j)}{H_i}$$

Own-opportunity reference

the reference prospect keeps the household's own jobs, availability and preferences, and equalises pay

Weighted generalised mean

the metric is a generalised mean of consumption with exponent β_c , weighted by leisure value and availability

the reference moves with the household's opportunity environment

Backup: two aggregation orders

Attained bundle

Gini within each simulated realisation, then averaged

$$I^{\text{ATT}}(S) = \frac{1}{R} \sum_r \mathcal{G}\left(\{M_{i,r}^{\text{ATT}}(T_S x_i)\}_i\right)$$

r : simulated realisation of attained bundles, common random numbers across coalitions

Ex-ante prospect

inverted within each household, then one Gini

$$I^{\text{EA}}(S) = \mathcal{G}\left(\{M_i^{\text{EA}}(T_S x_i)\}_i\right)$$

$M_i^{\text{EA}} = \lambda_c \exp\{(\log J_i - \log H_i)/\beta_c\}$, integrated over the household's prospect

the average of realised inequality versus the inequality of household equivalents

Backup: equivalisation

$$\tilde{M}_i^p = \frac{M_i^p}{e_i}, \quad e_i : \text{modified OECD equivalence scale}$$

Couples, ex ante: access + earnings

21.3% $\longrightarrow$ 7.9%

raw $\rightarrow$ equivalised; % of baseline Gini

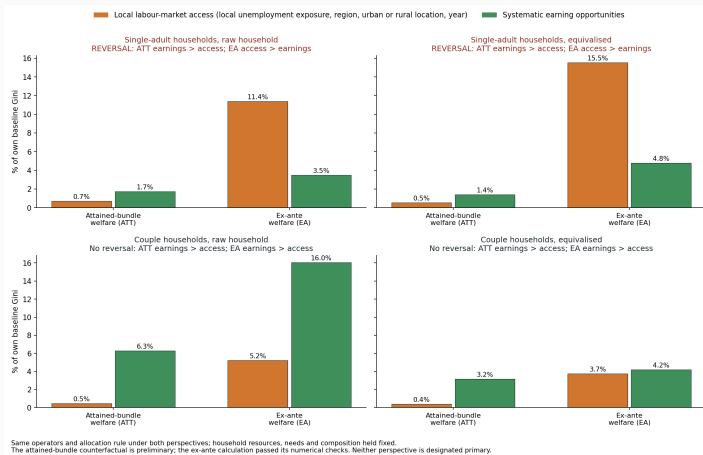
Couples, ex ante: preferences

0.00291 $\longrightarrow$ -0.00212

Gini points, raw $\rightarrow$ equivalised

a normative convention, reported both ways; populations never combined or compared

Backup: central figure



Backup: ex-ante decomposition in Gini points

	scale	baseline Gini	P	A	B	A + B, % of baseline	ordering
Single adults	raw	0.1595	0.00157	0.01812	0.00555	14.8	access > earnings
	equivalised	0.1406	0.00020	0.02183	0.00673	20.3	access > earnings
Couples	raw	0.1277	0.00291	0.00667	0.02049	21.3	earnings > access
	equivalised	0.1200	-0.00212	0.00449	0.00504	7.9	earnings > access

exact Shapley contributions in Gini points; household-weighted

Backup: attained-bundle decomposition in Gini points

	scale	baseline Gini	P	A	B	A + B, % of baseline	ordering
Single adults	raw	0.2337	0.0012	0.0016	0.0040	2.4	earnings > access
	equivalised	0.2451	-0.0003	0.0013	0.0034	1.9	earnings > access
Couples	raw	0.2035	0.0064	0.0009	0.0128	6.7	earnings > access
	equivalised	0.1971	-0.0019	0.0007	0.0062	3.5	earnings > access

exact Shapley contributions in Gini points; preliminary; household-weighted

Backup: ex-ante channels, % of baseline

		P				A		B	
	scale	I_0	Gini pts	% of I_0	Gini pts	% of I_0	Gini pts	% of I_0	
Single adults	raw	0.1595	0.00157	0.99	0.01812	11.37	0.00555	3.48	
	equivalised	0.1406	0.00020	0.14	0.02183	15.52	0.00673	4.79	
Couples	raw	0.1277	0.00291	2.28	0.00667	5.22	0.02049	16.04	
	equivalised	0.1200	−0.00212	−1.76	0.00449	3.75	0.00504	4.20	

each contribution divided by its own baseline Gini I_0 ; signed; the rows are not forced to sum to one hundred, because the remainder is outside the current P-A-B decomposition

Backup: attained-bundle channels, % of baseline

		P			A		B	
	scale	I_0	Gini pts	% of I_0	Gini pts	% of I_0	Gini pts	% of I_0
Single adults	raw	0.2337	0.0012	0.53	0.0016	0.69	0.0040	1.73
	equivalised	0.2451	-0.0003	-0.13	0.0013	0.54	0.0034	1.40
Couples	raw	0.2035	0.0064	3.16	0.0009	0.45	0.0128	6.28
	equivalised	0.1971	-0.0019	-0.98	0.0007	0.37	0.0062	3.15

each contribution divided by its own baseline Gini I_0 ; signed; preliminary

Backup: composition of the explained component

$$s_k = \frac{\phi_k}{\phi_P + \phi_A + \phi_B}, \quad k \in \{P, A, B\}$$

ex-ante prospect, equivalised	Single adults	Couples
Preferences s_P	0.7%	−28.5%
Local access s_A	75.9%	60.6%
Earning opportunities s_B	23.4%	67.9%
Total	100.0%	100.0%

These are shares of the currently explained P-A-B change, not shares of total inequality. Negative components are possible under Shapley attribution.

Backup: Shapley versus one-factor equalisation

$$\text{one-factor: } v^P(\{k\}) = I^P(\emptyset) - I^P(\{k\})$$

$$\text{Shapley: } \phi_k^P = \text{average over all orders of } v^P(S \cup \{k\}) - v^P(S)$$

$$\phi_P^P + \phi_A^P + \phi_B^P = I^P(\emptyset) - I^P(\{P, A, B\})$$

one-factor effects need not add up · Shapley contributions do, exactly

Backup: outside the current P-A-B decomposition

Held fixed

household resources, needs and
composition

Retained

sex-specific and
household-type-specific preference
and opportunity blocks

Not equalised

wage-draw luck and the common
offer spread

wage-offer location differences: at most about 0.10 log points

common offer spread: about 0.37 log points

next: extend the decomposition to household resources and needs, after fixing the appropriate operator

Backup: robustness of the attained-bundle decomposition

Second simulation seed

every coalition Gini and contribution
reproduces closely, including the preference
sign change

Observed job in the simulation

attained under every coalition by 2.9% of single
adults and 3.9% of couples
excluding it moves the total change by at most
9.3%, with no sign change

simulation checks, not confidence intervals

Backup: sample selection

screen	single-adult	couple
the raw France 2016 input file	4,038	5,965
one- or two-adult households (composition screen)	4,038	5,965
every adult aged 20 to 60	2,131	3,662
no adult still in education	2,036	3,521
no old-age, disability or survivor pension receipt	1,755	3,218
labour status in scope	1,598	2,412
no other employable or earning member	1,564	2,323
hours and wage inside the modelled support	1,555	2,275
hours outside [5,70] or unsupported military occupation (ISCO 0) on an employed decider	1,543	2,223
observed chosen alternative priced at non-positive disposable consumption	1,540	2,223
estimation sample	1,540	2,223

Backup: EUROMOD pricing

Inputs

a work arrangement: hours, hourly wage, earnings split; household roster and non-labour inputs

EUROMOD

French 2015 policy system; only deciders receive counterfactual overrides

Output

$C_i(j)$: household disposable income, summed over all members

every sampled alternative and every welfare integration point is priced
tax-benefit accounting identity holds to machine precision

Estimation

101

rows per household: the observed
job and 100 sampled alternatives

Predictive fit

2,048

common integration points per
household

Supports

[5, 70]

weekly hours; wages in [2, 590] per
hour

sampling devices are computational: they are not offers and enter no welfare measure